Avoral

Effervescent Granules

Urinary Alkalinizer

Composition:
Each sachet contains:

Sodium Bicarbonate	1.76g
Sodium Citrate	0.63g
Citric Acid	0.72g
Tartaric Acid	0.89g

Product Description:
White effervescent granules with lemon lime flavor, produce clear to slightly cloudy solution when dissolved in water. Sugar free and artificial sweetener free. Available in pack size of 12 sachets. Each sachet containing 4g of effervescent granules.

Indications:
Avoral is indicated for the relieve of discomfort caused by mild urinary tract infections (UTI), in which provides symptomatic relief of dysuria. It may enhance the action of certain antibiotics especially sulphonamide, and helps to prevent crystallisation of urates in gout especially when treated with uricosurics and possibly allopurinol. This alkalizer can be used for relieving symptoms associated with gastric hyperacidity.

Pharmacodynamics:
Sodium Bicarbonate acts as alkalinizing agent which increases the excretion of free bicarbonate ions in the urine, and thus raises the urinary pH. A rise in urinary pH increases the solubility of cysteine in the urine. By maintaining alkaline urine, the actual dissolution of uric acid stones may be accomplished. Sodium Bicarbonates acts as antacid as well by neutralizing or buffering existing quantities of stomach acid but has no direct effect in its output.

Sodium Citrate is metabolized to bicarbonates after absorption, thus providing similar pharmacological effect as per Sodium Bicarbonate. Citric Acid and Tartaric Acid both having transient effect as neutralizing or buffering agents.

Pharmacokinetics:
After absorption, Sodium Citrate is metabolized to bicarbonates. Sodium Bicarbonate is excreted through kidney and also via lung by forming carbon dioxide. Sodium Citrate, Citric Acid and Tartaric Acid are excreted through urine. Less than 5% of the citrates are excreted in the urine unchanged.

Dosage And Administration:
For oral use only. Dissolve 1-2 sachets in a glass (200ml) of water. Four times daily or as prescribed.

Contraindications:
Contraindicated in patients with severe renal impairment or renal failure, hypernatremia, metabolic or respiratory alkalosis, hypocalcaemia, hypochlorhydria, Addison's disease, acute dehydration, heat cramps, overt and occult cardiac failure and severe myocardial damage.
Not recommended in concomitant use with hexamine mandelate or hexamine hippurate therapy because an acidic urine is needed.
Avoid concomitant use with quinolone antibiotics as to prevent crystalluria in alkaline urine.

Warnings And Precautions:
Should not be taken by patients on a sodium restricted diet.
Caution should be taken in patients with peptic ulceration and/or renal abnormalities as to avoid the condition of metabolic alkalosis.
Should be given extremely cautiously to patients with heart failure, oedema, renal impairment, hypertension, eclampsia, and or aldosteronism.

Drug Interactions:
Use of urinary alkalizer theoretically may result in decreased therapeutic effect of chlorpropamide, lithium, salicylates and tetracyclines; increased therapeutic effect of amphetamines, ephedrine and pseudoephedrine.

Antacid: Concurrent use of antacids with Sodium Citrate and Sodium Bicarbonate may promote development of calcium stones in patients with uric acid stones and may also cause hypernatremia. Concurrent use of aluminium containing antacids with salts can increase aluminium absorption, possibly resulting in acute aluminium toxicity, especially in patients with renal insufficiency.
Quinolones: Citrates may reduce the solubility of ciprofloxacin, norfloxacin, or ofloxacin in the urine. Patients should be observed for signs of crystalluria and nephrotoxicity.

Pregnancy And Lactation:
Use in pregnancy and during lactation should be considered only when the possible benefits outweigh the potential risks.

Side Effects:
Sodium Bicarbonate given orally can cause stomach cramps, belching, and flatulence. Rarely, spontaneous rupture of the stomach has been reported.
Tartrate component of this product may not be completely absorbed. Due to this, it may exert mild laxative effect. Prolonged and excessive use may cause a systemic alkalosis and or hypernatremia.

Symptoms And Treatment For Overdosage:
Excessive use of bicarbonate or bicarbonate-forming compounds may lead to hypokalaemia and metabolic alkalosis, especially in patients with impaired renal function. Symptoms include mood changes, tiredness, slow breathing, muscle weakness, and irregular heart-beat. Muscle hypertonicity, twitching, and tetany may develop, especially in hypocalcaemic patients.
Treatment of metabolic alkalosis associated with bicarbonate overdose consists mainly of appropriate correction of fluid and electrolyte balance. Replacement of calcium, chloride, and potassium ions may be of particular importance.

Storage Condition:
Store below 30°C. Protect from moisture.

Shelf Life:
The expiry date is indicated on the packaging.

Product Owner
Xepa-Soul Pattinson (Malaysia) Sdn Bhd
1-5 Cheng Industrial Estate 75250 Melaka, Malaysia.

Product Registration Holder and Distributor
Apex Pharmacy Marketing Sdn Bhd
No.2, Jalan SS13/5, 47500 Subang Jaya, Selangor, Malaysia.

Contract Manufactured by:
Unison Nutraceuticals Sdn Bhd
No.13, Jalan Tasik Utama 52, Kawasan Perindustrian Tasik Utama, 75450 Ayer Keroh, Melaka.

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